

Qiuyi Wu

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		LinkedIn	linkedin.com/in/qiuyi-wu
Research Interests	Functional Data Analysis, Multi-Omics Integration, Biomedical Data Science, Text Mining, High Dimensional Regression, Recommender System, Image Processing, Spatial Statistics		
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Employment

Sep.2024 - Present	Duke University , Department of Biostatistics and Bioinformatics, US <i>Postdoc Research Fellow</i> Mentor: Dr. Anru Zhang & Dr. Pixu Shi
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Education

2019-2024	Ph.D., Statistics, United States University of Rochester , Department of Biostatistics and Computational Biology Advisor: Dr. Xing Qiu
2016-2018	M.S., Applied Statistics, United States Rochester Institute of Technology , School of Mathematical Sciences Advisor: Dr. Ernest Fokoue
2012-2016	B.Econ., Finance, China Donghua University , Glorious Sun School of Business and Management

Honors and Awards

Scholarship

- **William Jackson Hall Graduate Student Fellowship** for academic excellence, 2023-2024
awarded annually to one PhD student in UR Biostat Dept through the combination of outstanding performance in coursework and qualifying exams; excellence in their service as a graduate student teaching assistant; and timely completion of a dissertation containing work judged to be of particular significance in both its methodological contribution and potential impact in applications.
- **Dean's Ph.D. Fellowship** for PhD study in University of Rochester, 2019-2021
- **RIT Merit Scholarship** for Master Study in RIT, 2016-2018

Conference Award

- **Clinton Miller Award** for Best Student Poster in SRCOS Summer Research Conference, *Clemson*, 2024
- **Gold Medal for Best Methodology and Theory Award** in UPSTAT Conference, 2024
- **Gold Medal for Data Analytics Competition** in UPSTAT Conference, 2021
- **ASA Section on Text Analysis Best Student Presentation Award** in JSM Conference, 2020
- **Gold Medal for Best Student Research Award** in UPSTAT Conference, 2019

- **NC ASA Young Researcher Award** in AISC Conference, 2018
- **Gold Medal for Best Computation and Application Award** in UPSTAT Conference, 2018
- **Promising Academic Leader Award** in UPSTAT Conference, 2018
- **Young Scientific Leader Award** in UPSTAT Conference, 2017
- **Student Scholarship Winners** for SAS Global Forum, 2017

Travel Award

- **CWS Do-Bui Travel Award** for JSM conference, *Nashville*, 2025
- **ASA Early-Career Travel Award** for JSM conference, *Nashville*, 2025
- **ASA SSPA Conference Grant** for JSM conference, *Nashville*, 2025
- **IMSI NSF Travel Fund** for “Statistics Meets Tensors” Workshop, *Chicago*, 2025
- **ASA Travel Award** for Preparing to Teach Statistics and Data Science Workshop, *Oregon*, 2024
- **Travel Award** for FutureBAProf Workshop, *Iowa*, 2024
- **NSF/Boyd Harshbarger Travel Award** for SRCOS Summer Research Conference, *Clemson*, 2024
- **SIAM Travel Award** for SIAM SEAS Conference, *VT*, 2023
- **ASA BIPOC DWMP Travel Award** for JSM Conference, *Washington DC*, 2022
- **SAMSI Travel Award** for Closing Ceremony, *Duke*, 2021
- **ACNN Travel Award** for Big Data Neuroscience Workshop, *UMich*, 2019
- **SAMSI Travel Award** for Deep Learning Opening Workshop, *Duke*, 2019
- **ASA Travel Award** for IMS/ASA Spring Research Conference, *VT*, 2019
- **RIT Research & Creativity Travel Award** for JSM Conference, *Vancouver*, 2018

Publications

Published or Submitted:

- 2026** **Wu, Q.**, Zhu, Z., Zhang, A. (2026). Statistical Inference for Fuzzy Clustering (*submitted to JASA*)
- 2025** Zoe, D., Hannah, M., **Wu, Q.** et al (2025). “Do measures of general versus pregnancy-related anxiety have distinct maternal-fetal-placental biology?” *Journal of Affective Disorders*. doi: 10.1016/j.jad.2025.120673
- 2024** LeBlanc, P., Banks, D., Fu, L., Li, M., Tang, Z., **Wu, Q.** (2024). “Recommender Systems: A Review” *Journal of the American Statistical Association* (2024): 1-21. doi: 10.1080/01621459.2023.2279695
- 2023** Panisch, L., Murphy, H., **Wu, Q.** et al (2023). “Adverse childhood experiences predict diurnal cortisol throughout gestation.” *Psychosomatic Medicine*. 85(6), 507-516.
- 2022** **Wu, Q.** et al (2022). “A conditional approach for joint estimation of wind speed and direction under future climates.” *Advances in Statistical Climatology, Meteorology and Oceanography* 8.2: 205-224.
- 2022** Murphy, H., Gu, Y., **Wu, Q.**, et al (2022). “Prenatal Diurnal Cortisol: Normative Patterns and Associations with Affective Symptoms and Stress.” *Psychoneuroendocrinology* (2022): 105856.
- 2021** Frigau, L., **Wu, Q.**, Banks, D. (2021). “Optimizing the JSM Program.” *Journal of the American Statistical Association* (2021): 1-21. doi: 10.1080/01621459.2021.1978466
- 2018** **Wu, Q.**, Fokoue, E., & Kudithipudi, D. (2018). “An Ensemble Learning Approach to the Predictive Stability of Echo State Networks. *Journal Of Informatics And Mathematical Sciences*”, 10(1 & 2), 181 - 199. doi: 10.26713/jims.v10i1-2.827
- 2018** **Wu, Q.** (2018). “Statistical Aspects of Music Mining: Naive Dictionary Representation.” *Thesis*. RIT Scholar Works. Accessed from <https://scholarworks.rit.edu/theses/9932>

In Preparation:

- 2026** **Wu, Q.**, Shi, P. et al. (2026). Spatially Weighted Canonical Correlation Analysis of Cortical Metabolome and Proteome in the Aging Macaque Brain
- 2026** **Wu, Q.**, Qiu, X. (2026). Image Processing with Optimally Designed Parabolic Partial Differential Equations
- 2026** **Wu, Q.**, Qiu, X. (2026). Optimal Gaussian Kernel Smoothing Strategy
- 2026** Sadia, C., Nashae, P., Nafisa, I., Ying, M., **Wu, Q.** et al (2026). Prenatal Perfluoroalkyl Substances (PFAS) Alter Infant Cortisol Reactivity (submitted to Developmental Psychobiology)
- 2026** Nafisa, I., Sally, T., Donald, H., Yihui, G., Xing, Q., **Wu, Q.** et al (2026). Prenatal exposure to PM2.5 and NO2 and its association with child cortisol measures during the first year of life (submitted to ISEE 2025)
- 2026** **Wu, Q.** et al. (2026). Two book chapters in *AI for Purpose: Revolutionizing Healthcare Through Data Science*. (Eds. [K. Zou, L. DeTora, & D. Patel,]) Taylor & Francis, CRC Press. (Forthcoming):
 • Chapter 14: *Harmonizing Data and Creativity: Text Mining Techniques for Music Discovery in the Age of Generative AI*

Ad-hoc

- 2025** 40-minute Interview featured in THE NEXT STEP: Career and Grad School Advice from SMD Alumni
 “Finding a Path in Data and Discovery with Qiuyi Wu” [Podcast] [Youtube] [Spotify]
- 2021** Short Interview Piece appears in July 2021 issue of Amstat News magazine
 “TAIG Contest Winners Tell of Experience”

Research Experience

- Sep.2024 - Present** **Duke University**, Department of Biostatistics and Bioinformatics, US
Postdoc
 Mentor: Dr. Anru Zhang
 Research Fields: Post-clustering Inference; EHR; Translational Science
- Sep.2024 - Present** **Duke University**, Department of Biostatistics and Bioinformatics, US
Postdoc
 Mentor: Dr. Pixu Shi
 Research Fields: Spatial Multi-Omics Analysis
- Jun.2020 - Aug.2024** **University of Rochester**, Biostatistics and Computational Biology, US
PhD student
 Mentor: Dr. Xing Qiu
 Research Fields: Kernel Smoothing; Functional Data Analysis; Image Processing
- Jun.2022 - Aug.2022** **Liberty Mutual Insurance Company**, GRS Advanced Analytics team, US
Data Scientist Intern
 Mentor: Dr. Robert Yuen
 Project Topic: Predicting the profitability of construction projects via structured data and NLP
- Jun.2020 - Aug.2020** **Argonne National Laboratory (ANL)**, CELS/EVS Division, US
Research Aide Intern
 Mentor: Dr. Jiali Wang & Dr. Whitney Huang & Dr. Julie Bessac
 Research Fields: Uncertainty Quantification; Wind Modeling

- Jun.2019 - Aug.2019** **Argonne National Laboratory (ANL)**, CELS/MCS Division, US
Research Associate
 Mentor: Dr. Julie Bessac & Dr. Whitney Huang & Dr. Jiali Wang
 Research Fields: Spatial Statistics; Extreme Value Analysis
- May.2018 - May.2019** **Statistical and Applied Mathematical Sciences Institute (SAMSI)**, US
Graduate Research Fellow
 Mentor: Dr. David Banks
 Research Fields: Text Mining; Spatial Statistics
- Dec.2016 - Nov.2017** **Rochester Institute of Technology**, Department of Computer Engineering, US
Research Assistant
 Mentor: Dr. Dhireesha Kudithipudi & Dr. Ernest Fokoué
 Research Fields: Neural Networks; Ensemble Learning; Topic Modeling

Teaching Experience

- 2025** **Charles D. Owen Middle School:** Guest Scientist
- 3 Science classes [7th Grade] - Road To Science
- 2019-2023** **University of Rochester:** Teaching Assistant, Guest Lecturer
- BST 426 [Spr 24] - Linear Models
 - BST 426 [Spr 23] - Linear Models
 - BST 467 [Spr 21] - Applied Statistics in the Biomedical Sciences
 - BST 467 [Spr 20] - Applied Statistics in the Biomedical Sciences
- 2017-2018** **Rochester Institute of Technology:** Teaching Assistant
- STAT 747 [Spr 18] - Principles of Statistical Data Mining
 - MATH 251 [Spr 17] - Probability and Statistics

Consulting Experience

- Department of Biostatistics, University of Rochester.
 Responsibilities: To assist members of URMIC academic community with statistical design, data analysis, and software issues for their research.
- Environmental Influences on Child Health Outcomes (ECHO-UPSIDE project)
 Responsibilities: Statistical design, data analysis and paper writing for URMIC ECHO researches using models such as ICC, PCA, CCA, LMER etc.
 - Prenatal Diurnal Cortisol paper: I carried out the statistical analysis by building the lmer model, wrote code for the team and revised the statistical section of the paper.
 - Adverse Childhood Experience paper: I developed the analytic approach in the paper, edited and revised the final draft.
 - Fetal Growth project: I wrote a function to calculate estimated fetal weight percentile based on infant demographics and help the team gain insights about fetal growth curve patterns.
 - Immune Age Difference paper: I introduced the “immune age difference,” a novel variable reflecting the relative maturity of infants’ immune systems.
 - Baby Cortisol Analysis project: I conducted statistical imputation method to resolve the missing data issue in the analysis.
 - Mom Postnatal Analysis project: I did statistical analysis to batch correct the problematic batches and outliers in the data.

- Collaborate with Dr. Anita Disney and Michael Lutz, Duke University
Responsibilities: Developed statistical methods to investigate spatial proteomic and metabolomic organization in the aging primate brain.
- Collaborate with Dr. Miriam Treggiari, Duke University
Responsibilities: Focused on developing statistical strategies to understand how anesthesia and surgery affect postoperative brain waste clearance, sleep, and cognitive recovery.

Professional Service and Volunteerism

- **Volunteer Scientist**, *SciMatch for Middle Schools at NC Science Festival (SciFest)*, 2025
- **Treasurer**, *Research Triangle Regional Organization of Society for Risk Analysis (SRA)*, 2025 - Present
- **Senior Advisor**, *AI4Purpose Inc.*, 2024 - Present
- **Officer & Webmaster**, *ASA Section on Text Analysis*, 2023 - Present
- **Poster Session Organizer**, *From Foundations to Future of Informatics in Health Symposium*, 2025
- **Journal Reviewer**, *Harvard Data Science Review*, 2025
- **Journal Reviewer**, *Stat*, 2024
- **Journal Reviewer**, *Applied Stochastic Models in Business and Industry*, 2024
- **Session Chair**, "Approaches in Clustering for Analysis of Emerging Data Types", *JSM*, 2022
- **Committee Member**, *ICSA 2020 Applied Statistics Symposium Photo Contest*, 2021
- **Student Representative**, *ASA Text Analysis Interest Group*, 2020 - 2023
- **Event Editor**, *International Astrostatistic Association*, 2018 - 2020
- **Committee Member**, *ICSA 2020 Applied Statistics Symposium Talent Show*, 2020
- **Session Chair**, *8th Annual Conference of the Upstate New York Chapters of ASA*, 2019
- **Journal Reviewer**, *IEEE Transactions on Systems, Man and Cybernetics: Systems*, 2019
- **Session Organizer**, Chair Session "Application of Text Mining", *UpStat Conference*, 2018
- **Judge** for Pre-College Statistical Data Analysis Competition, *UpStat Conference*, 2018
- **Judge** for Undergraduate Data Competition, *ASA DataFest*, 2018
- **Mentor** for Data Competition, *ASA DataFest*, 2017 - 2018
- **Session Organizer**, Chair Session "Environment and Health", *UpStat Conference*, 2017
- **Technical Translator**, Deep Learning for NLP at Oxford with Deep Mind, *Big Data Digest*, 2017
- **Volunteer** for Imagine RIT: Innovation + Creativity Festival, *Imagine RIT*, 2017
- **Peer Mentor**, *CET Academic Programs*, 2015 (*Honored with Best Mentor via election*)

Mentoring

- **AI4Purpose Inc, Senior Advisor [2025-Present]**
*Advised a team developing an AI-driven digital health solution SomniaAI for Menopausal Insomnia:
Attended Versalio HITLAB Innovation Challenge in Oct 2024 and Swiss Venture Contest in Dec 2024;
Got invited to present in ASA's inaugural Artificial Intelligence (AI) Workshop NYC 2025;
Pitched to VCs and angel investors in Paradigm Talks in Aug 2025;
Filed a provisional patent application with the USPTO.* [App Demo]
- **Duke University, Informal Mentor [Summer 2025]**
Met with incoming undergraduate student to explore potential research directions prior to the start of first year.

Presentations

- Nov.2025** RIT Data Science Research Group Seminar - Rochester, US - DSRG2025
Invited, Soft Post-Clustering Inference with EHR Applications

- Oct.2025** Community Gust Lecture at Durham Center for Senior Life - Durham, US
Invited, Numbers That Matter: How Statistics Can Help You Make Smarter Decisions
- Aug.2025** Joint Statistical Meetings 2025 - Nashville, US - JSM2025
Topic Contributed, Soft Post-Clustering Inference with EHR Applications
- Apr.2025** Duke Industry Statistics Symposium - Durham, NC, US - DISS25
Poster, Image Processing with Optimally Designed Parabolic Partial Differential Equation
- Jan.2025** From Foundations to Future of Informatics in Health Symposium - Durham, NC, US - Ed90
Poster, Image Processing with Optimally Designed Parabolic Partial Differential Equation
- Aug.2024** Joint Statistical Meetings 2024 - Portland, OR, US - JSM24
Poster, Image Processing with Optimally Designed Parabolic Partial Differential Equation
- June.2024** 59th Southern Regional Council on Statistics Summer Research Conferenc - Clemson, SC, US - SRCOS24
Poster, Image Processing with Optimally Designed Parabolic Partial Differential Equation
- Apr.2024** 12th Annual Conference of the UPSTAT New York Chapters of ASA - Rochester, NY, US - UPSTAT24
Contributed, Image Processing with Optimally Designed Parabolic Partial Differential Equation
- Apr.2024** Johns Hopkins SMART WG seminar, Department of Biostatistics - Baltimore, MD, US - JobTalk
Invited, Image Processing with Optimally Designed Parabolic Partial Differential Equation
- Apr.2023** 11th Annual Conference of the UPSTAT New York Chapters of ASA - Rochester, NY, US - UPSTAT23
Contributed, A conditional approach for joint estimation of wind speed and direction under future climates
- Mar.2023** SIAM Southeastern Atlantic Section Conference 2023 - Blacksburg, VA, US - SIAM23
Contributed, A conditional approach for joint estimation of wind speed and direction under future climates
- Aug.2022** Joint Statistical Meetings 2022 - Washington DC, US - JSM2022
Topic Contributed, Text Mining and Music Mining
- May.2022** UR Biostatistics Department Annual Student Workshop - Rochester, NY - URMCM
Seminar Talk, Partial Differential Equation & Kernel Smoothing Modeling In Image Analysis
- Sep.2021** Classification and Data Analysis Group Conference - Florence, Italy - CLADAG
Invited, Minimizing Conflicts of Interest: Optimizing the JSM Program
- Aug.2021** Joint Statistical Meetings 2021 - Seattle (Virtual), US - JSM2021
Topic Contributed, Minimizing Conflicts of Interest: Optimizing the JSM Program
- Feb.2021** Data Science DC - Washington DC (Virtual), US - DSDC/TAIG
Invited, Text Mining and Music Mining [Video]
- Dec.2020** ICSA 2020 Applied Statistics Symposium - Texas (Virtual), US - ICSA2020
Poster, Bayesian and Unsupervised Machine Learning Machines for Jazz Music Analysis
- Aug.2020** SIAM Conference on Mathematics of Planet Earth - California (Virtual), US - MPE20
Contributed, Statistical Wind Conditions Assessment across inland and off-shore US under Future Climate Scenarios
- Aug.2020** Joint Statistical Meetings 2020 - Philadelphia (Virtual), US - JSM2020
Contributed, Naive Dictionary On Musical Corpora: From Knowledge Representation To Pattern Recognition
- Aug.2019** Argonne National Laboratory - ANL/MCS
Lightning Talk, Wind Conditions Assessment in North-America Under Climate Change Scenarios

- Jul.2019** Joint Statistical Meetings 2019 - Denver, US - JSM2019
Poster, Exploratory analysis of Hurricane Storm Surge
- May.2019** IMS/ASA Spring Research Conference - SRC 2019
Poster, Uncertainty Quantification in Tropical Cyclone Climatology
- May.2019** Statistical Perspectives on Uncertainty Quantification (SPUQ) Workshop - SPUQ 2019
Poster, Exploratory Analysis of Tropical Cyclone Climatology
- Apr.2019** 6th Bayesian, Fiducial, and Frequentist (BFF) Conferences - BFF 2019
Poster, Bayesian and Unsupervised Machine Learning Machines for Jazz Music Analysis
- Apr.2019** 8th Annual Conference of the UPSTAT New York Chapters of ASA - UpStat 2019
Poster, Text Mining and Music Mining
- Nov.2018** SAMSI Model Uncertainty Program Storm Surge Working Group - SAMSI
Seminar Talk, Initial Exploratory Analysis of Synthetic Storm Tracks (*30 minutes talk*)
- Oct.2018** SAMSI Model Uncertainty Program Data Fusion Working Group - SAMSI
Seminar Talk, Data Fusion for Music Mining (*40 minutes talk*)
- Oct.2018** International Conference on Advances in Interdisciplinary Statistics and Combinatorics - AISC
Contributed, Machine Learning for Music Mining with LDA Model, *SAMSI Academic Session*
- Sep.2018** Data Science Research Group in Rochester Institute of Technology - DSRG
Seminar Talk, Statistical Aspects of Music Mining (*80 minutes lecture*)
- Sep.2018** Cornell Day of Statistics 2018 - Cornell
Poster, Bayesian and Unsupervised Machine Learning Machines for Jazz Music Analysis
- Jul.2018** Joint Statistical Meetings 2018 - Vancouver, Canada - JSM
Poster, Bayesian and Unsupervised Machine Learning Machines for Jazz Music Analysis
- Apr.2018** 7th Annual Conference of the UPSTAT New York Chapters of ASA - UpStat 2018
Contributed, Music Mining In Topic Modeling Approach For Improvisational Learning
- Feb.2018** Graduate Seminar in Rochester Institute of Technology - RIT
Seminar Talk, Topic Modeling with LDA Tutorial (*80 minutes lecture*)
- Nov.2017** Graduate Showcase in Neuroscience and Signal Processing Session - RIT
Contributed, Statistical Challenges of Echo State Networks
- Nov.2017** 6th Annual Conference of the UPSTAT New York Chapters of ASA - UpStat 2017
Contributed, Statistical Aspects about Echo State Networks

Professional Affiliation

- American Statistical Association (ASA)
- Institute of Mathematical Statistics (IMS)
- International Astrostatistics Association (IAA)
- International Society for Bayesian Analysis (ISBA)
- Astrostatistics and Astroinformatics Portal (ASAIP)
- International Chinese Statistical Association (ICSA)
- Caucus for Women in Statistics and Data Science(CWS)

Skills and Interests

- **Programming:** Python, R, SAS, MATLAB, L^AT_EX, SQL
- **Marathon:** Finish 26.2-mile journey in limited time, 2017, 2018, 2019
- **Music:** Perform Ghanaian and Senegalese music as member of African Percussion & Dance Ensemble
- **Other:** Zumba, Piano, Hot Yoga, Surfing, Snowboarding, Hiking, Climbing, Taekwondo, Sketching, Culinary Arts